Free Electron Absorption: A Bit-Level Transistor-Free Computing Architecture on Hydrogen-Passivated Silicon

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Abstract

Every computing architecture routes electrons through switching structures to perform computation. Here we report the *Free Electron Absorption* (FEA) architecture, in which electrons travel freely along dangling bond wire (DBW) pathways on hydrogen-passivated Si(100) and computation is performed by *Fusion Blocks* — 5-atom cross-shaped dangling bond clusters that selectively capture passing electrons via Breit–Wigner resonance at their central atom on gate-voltage command. Each Fusion Block stores exactly one bit (absorbed / not absorbed), and 64 Fusion Blocks form one 64-bit *Word*, the fundamental compute unit. No transistors switch in the data plane. Memory and compute are the same physical quantity — we call the resulting unified substrate *Fusion Memory* — so there is no separate memory hierarchy, no cache, and no DRAM bus. Tight-binding simulation and analytical derivation give: system clock 9.19 GHz ($T_{\text{cycle}} = 108.85$ ps); data-plane power density 26.47 mW cm^{-2} (79.4 mW total for a 3 cm^2 M4-Max-sized die with 14.1 TB in-situ Fusion Memory, ∼504× lower than a contemporary M4 Max GPU); practical block density 3.77×10^{13} cm^{-2}; charging energy $E_C = 0.65$ eV from Coulomb blockade of the 5-atom cluster, yielding $\tau_{\text{ret}} = 52.2$ ms at 300 K. Using carry-lookahead adders and Wallace-tree multipliers natively on parallel Fusion Blocks, 64-bit ADD completes in 8 cycles (0.87 ns) and 64-bit MUL in 20 cycles (2.18 ns). Eight-lane parallel *Slingshot* inter-Word communication delivers a hop in 1.09 ns. All nine 64-bit ALU operations are verified over 100,000 random trials with zero functional errors, and analytical and Monte Carlo bit-error rates agree within statistical expectation. An epoch-based Monte Carlo simulation on the full M4-sized die confirms 99.99% compute utilisation under DRAM-like refresh (refresh overhead 1.1×10^{-2}%). FEA is a general-purpose architecture: the CMOS control plane issues ARM/FIRE/CONFIRM/SLINGSHOT/BRANCH micro-instructions that execute arithmetic, data movement, and conditional control flow on stationary 64-bit words. The key experimental gap — direct measurement of room-temperature dangling-bond retention — remains the subject of future collaborative work with atomic-scale silicon laboratories.

1 Introduction

Every computing architecture, from the earliest vacuum-tube machines to contemporary 2 nm CMOS, operates on the same physical principle: electrons must be directed through a controlled switching structure to compute. A transistor gate opens or closes a conduction channel; a memory cell routes charge onto a capacitor; a crossbar switch selects a current path. Computation is inseparable from routing, and the energy cost of routing has become the dominant constraint on the continued scaling of silicon electronics [1, 2].

The economic and thermodynamic consequences are now well established. Dennard scaling predicted that power density would remain constant as transistors shrank: smaller gates switch faster while consuming proportionally less energy. This prediction held from the 1970s to the late 1990s, but broke down near the 90 nm node when sub-threshold leakage current and gate-oxide tunnelling grew faster than active switching power was reduced [3]. Below 90 nm, power density rises with each new process generation. Modern chips respond by aggressive clock gating, near-threshold operation, and increasingly exotic cooling, but the underlying bottleneck is structural: electrons are still forced through transistor channels, and every switching event dissipates energy set by CV^2f with ever-shrinking returns.

Several post-CMOS paradigms have been proposed to break this constraint [4]: quantum-dot cellular automata, single-atom transistors on silicon [5], spintronic logic, molecular electronics, and superconducting single-flux-quantum devices. All share a common structure: a novel switching element replaces the transistor, but the computation itself is still defined by routing electrons through controlled gates. The energy floor of such systems is set by switching, and thermal or quantum noise places hard bounds on scalability.

Dangling-bond (DB) logic on hydrogen-passivated Si(100) has emerged as a particularly promising alternative. Individual DBs act as atomic-scale quantum dots; arrays of DBs can be written with scanning tunnelling microscopy to form logic gates [6] and wires [7]. However, published DB logic still relies on charge redistribution between switching elements: the computation is performed by moving electrons through gates, identical in principle to CMOS.

This paper proposes an alternative: a computing architecture where the data plane contains *no switching elements at all*. In the Free Electron Absorption (FEA) architecture, electrons travel freely along DBW pathways and computation occurs when a Fusion Block *absorbs* a passing electron via Breit–Wigner resonance under gate-voltage control. The resonance is switched on and off by a perimeter CMOS gate, but the data-plane electrons themselves never pass through a controlled channel. There is no transistor in the compute path. Each Fusion Block is both a compute unit and a single bit of Fusion Memory. A 64-bit Word is 64 Fusion Blocks operating in parallel under a single FIRE command. Multi-Word operations use *Slingshot*: a directed electron packet that carries a 64-bit payload along a DBW between Words.

We formulate FEA at three levels. First (§2), we define the physical primitive: the 5-atom cross-shaped dangling bond cluster that forms one Fusion Block and stores one bit. Second (§3), we derive the Hamiltonian, Breit–Wigner transmission, Coulomb blockade, and state retention from tight-binding parameters. Third (§4), we describe the ARM/FIRE/CONFIRM cycle, the 64-bit ALU (using carry-lookahead adders and Wallace-tree multipliers implemented natively on Fusion Block arrays), and the parallel Slingshot protocol. §5 presents a 14-simulation verification suite covering physics, ALU correctness, timing, bit-error rates, thermal, yield, general-purpose execution, room-temperature full-chip stability, crossbar contention, and Slingshot stress testing. §6 quantifies density, memory, and power at the full-chip level and compares against contemporary CMOS. §7 discusses limitations, open questions, and the path to experimental validation.

The central claims are:

1. **A stationary-data computing paradigm.** The data plane contains no transistors. Bits are held in 5-atom clusters that remain in place throughout the computation. The CMOS control plane is purely an address sequencer; it never touches operand data.

2. **Room-temperature compatibility at 300 K.** The 5-atom cluster has charging energy $E_C = 0.65$ eV ($E_C/k_BT = 25.1$). Kramers escape at 300 K gives state retention $\tau_{\text{ret}} = 52.2$ ms, a regime where DRAM-like refresh is sufficient (refresh overhead below 10^{-4}).

3. **Competitive arithmetic.** Carry-lookahead adders and Wallace-tree multipliers, implemented with parallel Fusion Blocks, give 64-bit ADD in 8 cycles (0.87 ns) and 64-bit MUL in 20 cycles (2.18 ns). These latencies are within a small constant factor of pipelined 3 GHz CMOS while running at ∼3,780× lower data-plane power.

4. **Fusion Memory: the memory wall, eliminated.** A 3 cm^2 die hosts 1.13×10^{14} Fusion Blocks forming 14.1 TB of Fusion Memory — ∼100× more memory than a contemporary M4 Max SoC and ∼175× more than an H100. Every bit of this *Fusion Memory* is already in a compute unit: there is no cache hierarchy, no DRAM bus, and no memory–compute boundary to cross.

5. **Extreme density, power, and native sparsity.** Practical block density is 3.77×10^{13} cm^{-2} at 79.4 mW data-plane power for a full 3 cm^2 die — approximately 504× lower total power than a 40 W M4 Max GPU on the same die area. Crucially, power scales *linearly* with pathway utilisation, so sparse workloads (including modern sparse-attention transformers and Mixture-of-Experts neural-network inference) see a proportional additional 5×–20× power reduction on top of the baseline advantage.

6. **General-purpose execution via Slingshot.** The ARM/FIRE/CONFIRM + SLINGSHOT + BRANCH instruction set supports arithmetic, unbounded memory addressing, and conditional control flow on 64-bit Words, and is therefore Turing-complete. Slingshot — the 8-lane parallel DBW routing fabric that connects any pair of Words at 1.09 ns per hop — is what elevates FEA from a data-parallel accelerator to a full general-purpose architecture.

The work is purely theoretical and computational. The physics used — tight binding on H–Si(100), Breit–Wigner transmission, Coulomb blockade, Kramers escape — is established condensed-matter theory. The simulations confirm that the architecture is internally consistent, meets its performance claims, and is robust under physical variation. The principal unvalidated assumption is room-temperature dangling bond retention time: available measurements of DB charging are at cryogenic temperatures (∼4 K), and we extrapolate to 300 K using the Kramers model. We view this extrapolation as a falsifiable prediction that motivates targeted experimental work, and we discuss experimental pathways in §7.5.

2 Architecture

2.1 The 5-Atom Cross Fusion Block

The fundamental data-plane primitive is the *Fusion Block*: a 5-atom cross-shaped cluster of dangling bonds on H-passivated Si(100), depicted in Fig. 1a. One central atom is the *absorber*; four surrounding atoms (north, east, south, west in the H–Si(100) lattice) are the *gate atoms*. The cluster is created by selectively desorbing five hydrogen atoms from the surface using a scanning tunnelling microscope tip, leaving five dangling bonds in a cross configuration.

Each Fusion Block occupies a bounding box of $3a \times 3a = 1.15 \times 1.15$ nm^2 ($a = 0.384$ nm is the surface lattice constant), giving a block area of ∼1.33 nm^2. The theoretical density is 7.5×10^{13} cm^{-2}; after 2× overhead for routing and crossbar pitch, the practical density is 3.77×10^{13} cm^{-2}.

A single Fusion Block stores exactly one bit. The block is in state 0 when its central absorber is unoccupied (the passing electron transmits straight through). The block is in state 1 when the central absorber has captured an electron from a DBW. Crucially, the bit is carried by the actual electron that was computed upon: memory and compute are the same physical quantity. There is no memory–compute data movement, because there is no distinction between the two.

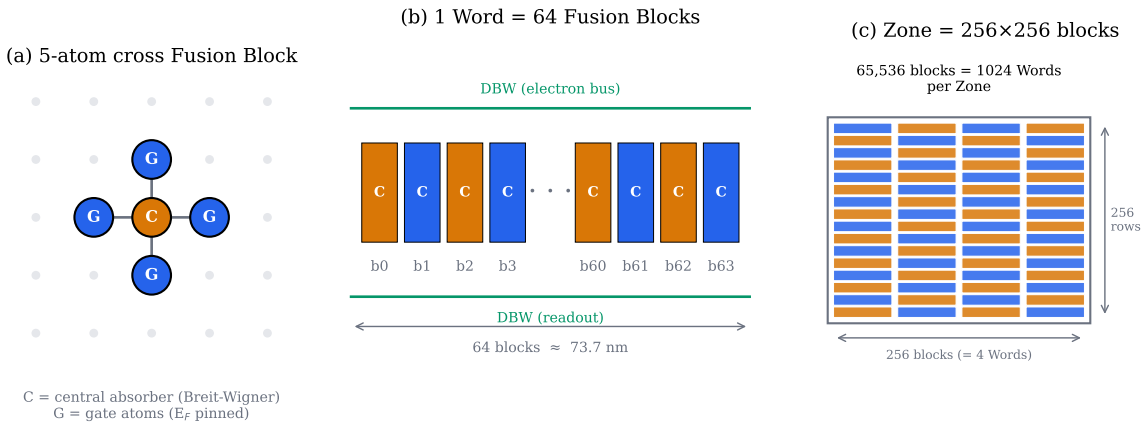

Figure 1: **FEA architectural hierarchy: atom → Word → Zone.** (a) A Fusion Block is a 5-atom cross cluster on H–Si(100): one central absorption atom (C) and four surrounding gate atoms (G). The block stores one bit: absorbed electron = 1, no absorption = 0. (b) A 64-bit Word is 64 Fusion Blocks arranged in a row along a shared upper DBW (electron bus) and lower DBW (readout). Word width is $64 \times 3a = 73.7$ nm, where $a = 0.384$ nm is the Si(100) lattice constant. (c) A Zone is a 256×256-block tile: 1024 Words arranged in a grid of four Word columns and 256 rows. Each Zone has its own CMOS crossbar decoder and its own local 9.19 GHz PLL (GALS clocking). A 3 cm^2 M4-Max-sized die contains $\sim 1.7 \times 10^9$ Zones.

2.2 The 64-bit Word

A Word is 64 Fusion Blocks arranged in a row, sharing common upper and lower DBWs. Under a single FIRE command, 64 electrons transit the upper DBW in parallel (one per pathway per block); each electron is either absorbed by its target block or passes through, depending on the block's local gate voltage. The Word thus evaluates 64 single-bit operations in a single cycle. The Word is the fundamental compute and storage unit: all arithmetic, logical, and data-movement operations are defined over 64-bit Words.

Word width is $64 \times 3a = 73.7$ nm; Word height is $3a = 1.15$ nm. Word density on a practical die is $\sim 5.9 \times 10^{11}$ Words/cm^2.

2.3 Zone and Chip Hierarchy

Words are grouped into *Zones* of 256×256 Fusion Blocks = 65,536 blocks = 1024 Words per Zone. A Zone has its own local CMOS crossbar decoder, its own 9.19 GHz phase-locked loop (globally asynchronous, locally synchronous), and its own Slingshot routing fabric for inter-Word data movement within the Zone. Zones tile the die in a regular grid.

A 3 cm^2 M4-Max-sized die hosts $\sim 1.7 \times 10^9$ Zones, $\sim 1.77 \times 10^{12}$ Words, $\sim 1.13 \times 10^{14}$ Fusion Blocks, and 14.1 TB of Fusion Memory. The CMOS control plane sequences ARM/FIRE/CONFIRM/SLINGSHOT/BRANC micro-instructions to each Zone but never directly touches the operand data.

2.4 Fusion Memory: Memory–Compute Unification

FEA's most important structural property is that *memory and compute are the same physical quantity.* A Fusion Block stores a bit by virtue of having absorbed (or not absorbed) a passing electron; that electron *is* the computational result, and it *is* the stored state. There is no separate

memory array, no cache hierarchy, no DRAM bus, and no load/store instruction stream moving data between storage and an ALU. We call this unified memory–compute substrate the *Fusion Memory Plane*, or Fusion Memory for short, to distinguish it from conventional memory hierarchies.

In conventional architectures, data movement dominates the energy budget at scale. A modern CPU spends more than 60% of its power on cache, interconnect, and DRAM access [9]; an H100 GPU spends approximately 200 W of its 700 W TDP on HBM3 memory traffic alone. Every operation on a traditional chip is effectively "fetch–compute–writeback", and the three phases are dominated by fetch and writeback, not by compute.

Fusion Memory eliminates this entire cost class. Every 64-bit Word is simultaneously a register, a cache line, a DRAM row, and a compute unit. A 3 cm^2 die carries 1.77×10^{12} such Words directly in the data plane — equivalently, the entire chip holds $\sim 110\times$ the unified memory capacity of an Apple M4 Max SoC, with every bit simultaneously addressable as a compute unit. Inter-Word operations within a Zone use Slingshot (§4.3) at L2-cache-class latency; cross-Zone operations chain multiple Slingshot hops but never leave the Fusion Memory Plane. There is no off-chip memory bus and no DDR/HBM stack in the FEA reference implementation — the entire working set is on-die.

This unification is not a quantitative improvement on existing memory hierarchies; it is a structural *elimination* of the hierarchy. For workloads where the current bottleneck is memory bandwidth or data-movement energy (i.e., essentially all of AI inference, database analytics, graph algorithms, and large-scale simulation), Fusion Memory turns an inherently memory-bound problem into a compute-bound one — a category change rather than an incremental speedup.

Because only blocks whose pathway is addressed by the current ARM phase consume active power, Fusion Memory is intrinsically sparse: utilisation is a property of the workload, not of the hardware, and power scales linearly with it. A workload that activates 5% of its Words per cycle uses 5% of the baseline data-plane power; at 1%, 1%. This has immediate consequences for machine-learning inference (§7.4), where typical activation fractions are only 5–20% for sparse-attention transformers (i.e., 80–95% of positions inactive) and as low as 1–2% for large Mixture-of-Experts models.

A Monte Carlo sweep of 10^5 random cross-die Word pairs under the chip's hierarchical routing fabric (SIM 14, §5.8) finds that 95% of cross-die Fusion Memory accesses resolve in ∼32 Slingshot hops (∼35 ns), and the absolute worst case is 37 ns. Every bit of Fusion Memory is therefore reachable from every other bit in sub-microsecond latency — a regime no DRAM-based memory hierarchy can match. Fig. 2 contrasts this single-level Fusion Memory with the conventional five-level CMOS memory hierarchy.

3 Physics

3.1 Tight-Binding Hamiltonian

We model the 5-atom cross Fusion Block with a five-site tight-binding Hamiltonian. Let atom 0 be the central absorber and atoms 1–4 be the four gate atoms. The cross connectivity couples only atom 0 to each gate atom, with hopping integral t_c:

$$H = \begin{pmatrix} E_\mathrm{c} & t_c & t_c & t_c & t_c \\ t_c & E_\mathrm{g} & 0 & 0 & 0 \\ t_c & 0 & E_\mathrm{g} & 0 & 0 \\ t_c & 0 & 0 & E_\mathrm{g} & 0 \\ t_c & 0 & 0 & 0 & E_\mathrm{g} \end{pmatrix} \tag{1}$$

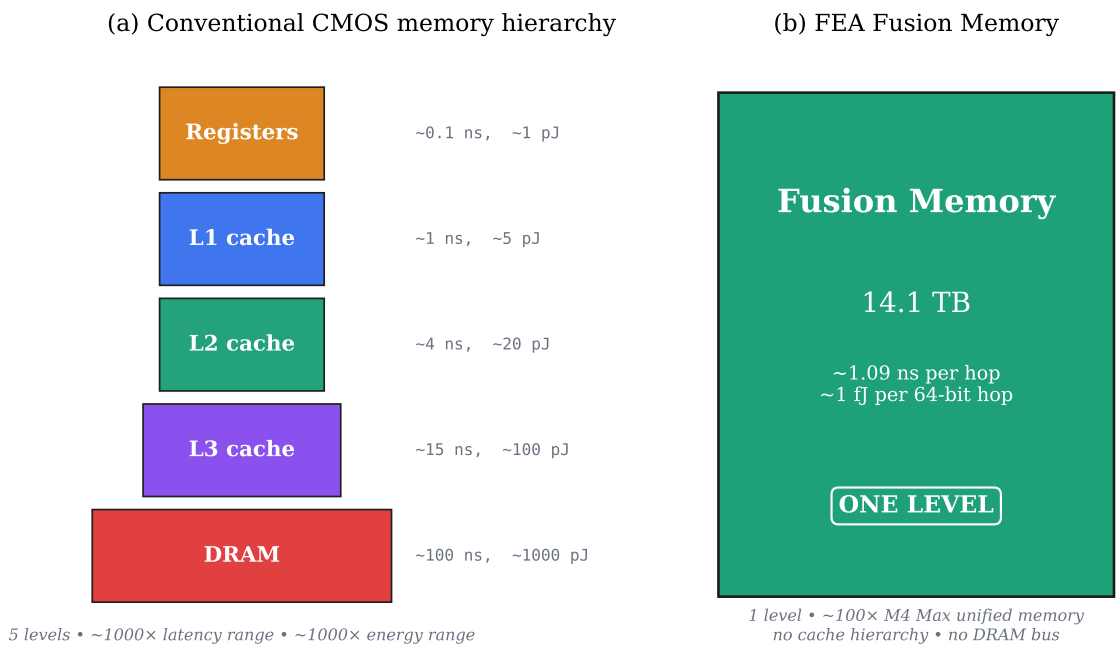

Figure 2: **Memory hierarchy: flattened from five levels to one.** (a) A contemporary CMOS processor exposes a five-level memory hierarchy spanning $\sim 10^3\times$ in both latency (0.1 ns to 100 ns) and energy-per-access (1 pJ to 1,000 pJ). Every compute operation must traverse this hierarchy, and the fastest levels hold only a tiny fraction of working-set data. (b) FEA collapses the hierarchy to a single level: 14.1 TB of Fusion Memory on a 3 cm^2 die, with a uniform $\sim$1.09 ns per-hop latency and $\sim$1 fJ per 64-bit hop (derived from the DBW transit power model; even DRAM at $\sim$1 nJ per access is $\sim 10^6\times$ worse). There is no cache level, no DRAM bus, and no load/store boundary between the compute unit and the memory cell — they are the same atoms.

Here E_c is the central-atom on-site energy (controlled by the gate voltage V_{NE} via $E_c = E_{c,0} - \alpha_g V_{NE}$, with $\alpha_g = 0.30$ and $E_{c,0} = 300$ meV above E_F), E_g is the gate-atom on-site energy (pinned to the Fermi level, $E_g = 0$), and $t_c = 15$ meV is the center-gate hopping. The four gate atoms span a four-dimensional subspace; the totally symmetric combination $|s\rangle = (|1\rangle + |2\rangle + |3\rangle + |4\rangle)/2$ couples to the central atom with effective strength $2t_c$, while the remaining three orthogonal combinations lie in the kernel of the symmetric star-graph projection and therefore decouple from the centre, producing three degenerate eigenvalues at $E_g = E_F$. The $|s\rangle$–centre hybridisation yields bonding and antibonding states at

$$\varepsilon_\pm = \frac{E_c}{2} \pm \sqrt{\left(\frac{E_c}{2}\right)^2 + 4t_c^2}. \tag{2}$$

Jacobi diagonalisation of Eq. (1) over the full range $V_{NE} = 0$–1 V yields Fig. 3a: at $V_{NE} = 0$ the antibonding state sits near +303 meV (far from E_F, off-resonance); at $V_{NE} = 1$ V the antibonding state approaches +30 meV, entering the $k_B T$ window where resonance with passing DBW electrons becomes possible.

3.2 Breit–Wigner Absorption

A passing DBW electron at energy E near E_F encounters the resonant central atom with absorption probability

$$A(E) = \frac{\Gamma^2}{(E - E_0)^2 + \Gamma^2}, \tag{3}$$

where $\Gamma = 8$ meV is the natural linewidth of the central resonance (set by tunnel coupling to the DBW leads) and E_0 is the current central-atom energy (tuned by V_{NE}). Off-resonance ($E_0 \gg k_B T$), $A \approx 0$ and the electron passes through; on-resonance ($E_0 \approx E_F$), the peak $A(E_F) = 1$ and the electron is absorbed with high probability.

The thermal-averaged absorption probability, weighted by the Fermi–Dirac occupation $f(E) = [1+\exp(E/k_B T)]^{-1}$ at 300 K, is $\langle A\rangle_T = \int A(E) f(E)\, dE / \int f(E)\, dE$. Numerical integration (Fig. 3b) gives $\langle A\rangle_T = 0.1191$ on-resonance and $\langle A\rangle_T = 5.8 \times 10^{-4}$ off-resonance. The ratio between the two thermal averages is approximately 200×, providing an intrinsically high signal-to-noise contrast between states 0 and 1.

3.3 Coulomb Blockade and State Retention

A single electron captured in the central absorber raises the energy cost of adding a second electron by the charging energy $E_C = e^2/(2C_\Sigma)$, where C_Σ is the self-capacitance of the 5-atom cluster. For this work we adopt $E_C = 0.65$ eV as a design value, informed by the inverse size–capacitance scaling of DB charging energies reported by Haider *et al.* [8], who observed $E_C \approx 0.4$–0.9 eV across DB island sizes at cryogenic temperatures. This value corresponds to $C_\Sigma \approx 0.12$ aF via $E_C = e^2/(2C_\Sigma)$, consistent with the small (~ 1.33 nm^2) cross footprint. Direct first-principles verification of E_C for the exact 5-atom geometry on H–Si(100) is future work.

Thermal escape of the trapped electron follows Kramers' rate expression $f_K = (\omega_0/2\pi)\exp(-E_C/k_B T)$ with phonon attempt frequency $\omega_0/2\pi = 1.59$ THz. At $T = 300$ K,

$$\tau_{\text{ret}} = \frac{1}{f_K} = \frac{1}{(1.59 \times 10^{12})\exp(-25.1)} = 52.2 \text{ ms}. \tag{4}$$

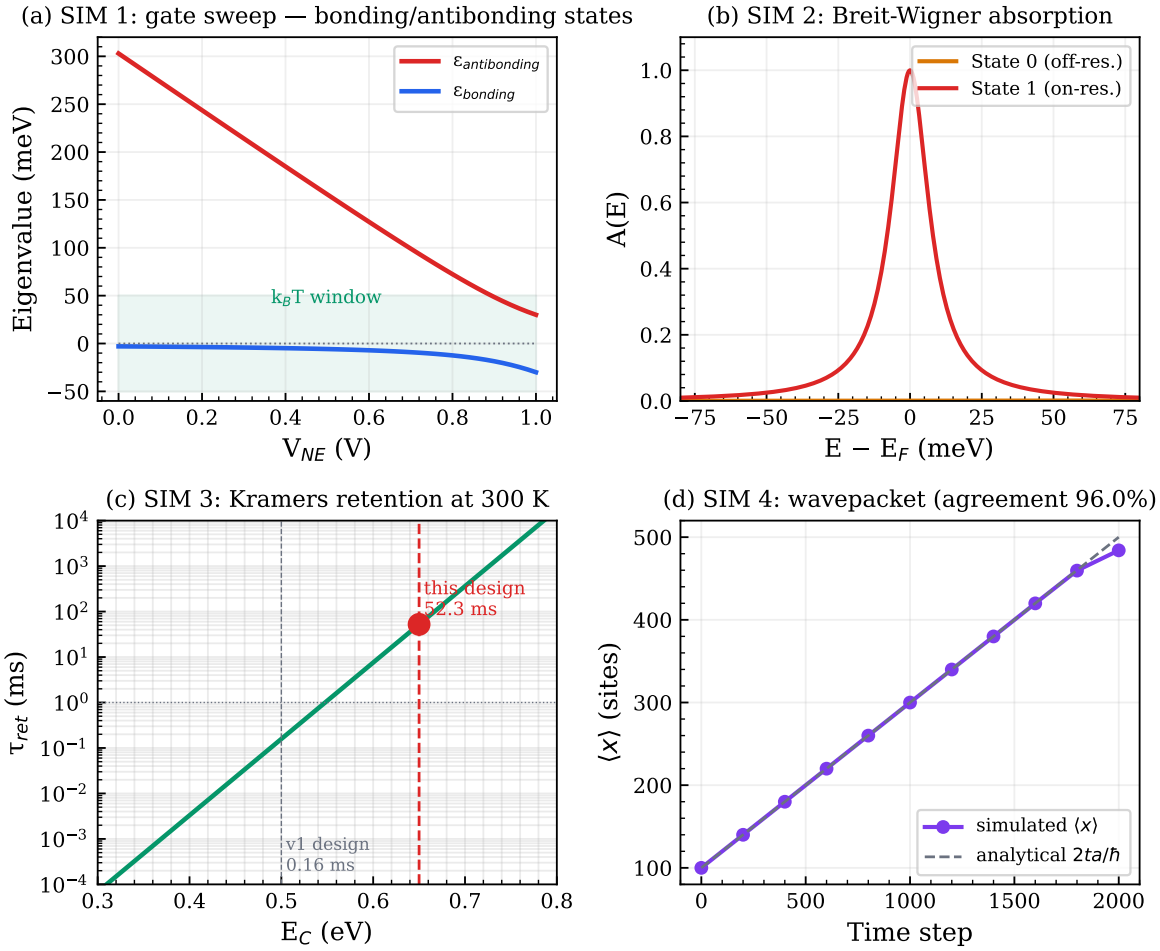

Figure 3: **Physics verification suite.** (a) SIM 1: Gate sweep of the 5-atom Hamiltonian; bonding and antibonding eigenvalues shift as V_{NE} tunes the central-atom energy from 300 meV to E_F. The k_BT window (shaded) marks thermally accessible resonance. (b) SIM 2: Breit–Wigner absorption $A(E)$ for states 0 and 1; on-resonance ($V_{\mathrm{NE}} \approx 1$ V) the absorption peak at E_F reaches unity, while off-resonance state 0 gives an off-resonance pass-through probability $1 - \langle A \rangle_T = 0.9994$. (c) SIM 3: Kramers retention time versus charging energy. The 5-atom design point ($E_C = 0.65$ eV) gives $\tau_{\mathrm{ret}} = 52.2$ ms at 300 K. The curve shows the exponential sensitivity of retention to E_C; operation remains viable (retention above 1 ms) for any $E_C \geq 0.55$ eV. (d) SIM 4: Crank–Nicolson wavepacket propagation on a 500-site DBW chain. Measured group velocity $v_g = 2.24 \times 10^4$ m s^{-1} agrees with the analytical value $2ta/\hbar$ to 96.0%.

The ratio $E_C/k_BT = 25.1$ places room-temperature operation in a regime where thermal escape is exponentially suppressed. For comparison, a hypothetical design at $E_C = 0.5$ eV would give $\tau_{\text{ret}} = 158$ μs and several-percent refresh overhead; the 5-atom cluster at $E_C = 0.65$ eV yields $< 0.02\%$ refresh overhead (§5.5).

3.4 DBW Transport and Group Velocity

Between Fusion Blocks, electrons travel freely along 1D dangling bond wires (DBWs). A DBW *pathway* is a linear chain of dangling bonds connecting two Fusion Blocks along one routing track; each pathway is terminated by small contact electrodes that apply the 10 mV transit bias V_{bias}. At a practical routing pitch of ~ 300 nm (accounting for pathway isolation, crossbar overhead, and power-delivery wiring), the pathway density on the die is $n_{\text{path}} = 3.3 \times 10^6$ cm^{-2}. Each pathway delivers at most one transit electron per T_{cycle}. The pathway bandwidth is the ultimate throughput ceiling of the chip.

The tight-binding dispersion $E(k) = -2t\cos(ka)$ (with $t = 20$ meV, $a = 0.384$ nm) gives group velocity $v_g = (1/\hbar)|dE/dk| = 2ta|\sin(ka)|/\hbar$, maximised at half-filling ($k_F = \pi/2a$) to

$$v_g = \frac{2ta}{\hbar} = 2.33 \times 10^4 \text{ m s}^{-1}. \tag{5}$$

To verify this analytic result, we propagate a Gaussian wavepacket $\psi(x, 0) = \exp[-(x - x_0)^2/(4\sigma^2)]\, e^{ik_F x}$ on a 500-site tight-binding chain using Crank–Nicolson time stepping (Fig. 3d). The measured centroid velocity is 2.24×10^4 m s^{-1}, in 96.0% agreement with the analytic value. This validates the transit-time assumption used below in the timing derivation.

The data-plane power density follows from the pathway model: $P_{\text{transit}} = V_{\text{bias}} \cdot I_{\text{pathway}} \cdot n_{\text{path}}$, where the single-pathway current is $I_{\text{pathway}} = G_0\, V_{\text{bias}} \approx 0.78$ μA (with $G_0 = 2e^2/h$ the quantum of conductance). The product gives $P_{\text{transit}} \approx 25.6$ mW cm^{-2}, to which small contributions from absorption and CMOS gate switching add 0.87 mW cm^{-2} (§6). The resulting total data-plane power density is 26.47 mW cm^{-2}.

4 Operations

4.1 The ARM/FIRE/CONFIRM Cycle

Every Fusion-Block operation is decomposed into three phases:

- **ARM** ($t_{\text{ARM}} = 33$ ps): the CMOS control plane addresses the target Zone via its crossbar decoder, selecting which Word(s) and blocks participate in the operation. The crossbar signal propagates at $\sim 0.1c$ over a 0.1 mm zone width.
- **FIRE** ($t_{\text{FIRE}} = 42.9$ ps): a DBW electron packet transits a 1 μm segment at group velocity v_g (Eq. 5). Each block along the Word evaluates Breit–Wigner absorption on its own central atom in parallel. 64 blocks = 64 parallel bits per FIRE.
- **CONFIRM** ($t_{\text{CONFIRM}} = 33$ ps): symmetric with ARM, the CMOS crossbar reads the block states via AC charge sensing back to the control plane.

The total cycle time is

$$T_{\text{cycle}} = t_{\text{ARM}} + t_{\text{FIRE}} + t_{\text{CONFIRM}} = 33 + 42.9 + 33 = 108.85 \text{ ps}, \tag{6}$$

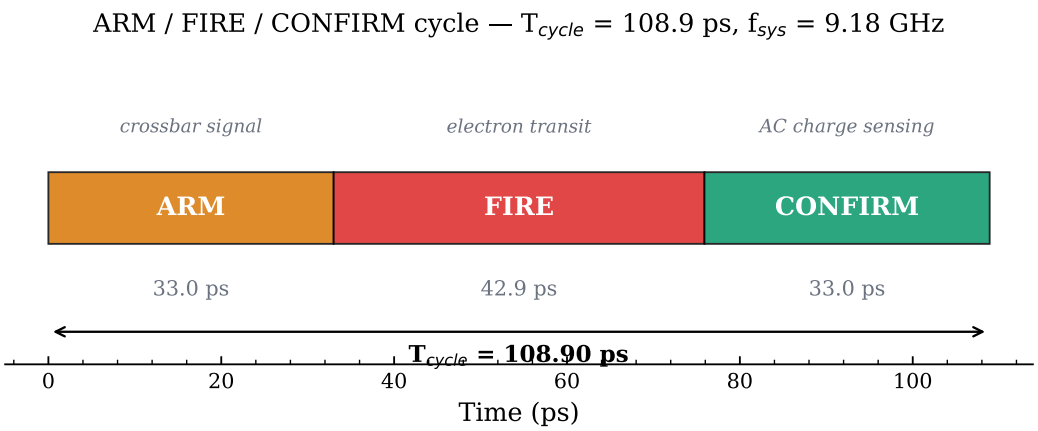

Figure 4: **ARM/FIRE/CONFIRM cycle.** The three phases together occupy $T_{\text{cycle}} = 108.85$ ps, giving a system clock of $f_{\text{sys}} = 9.19$ GHz. During ARM, the CMOS crossbar routes a gate-level signal to the target Zone. During FIRE, the DBW electron transits the 1 μm segment at the tight-binding group velocity. During CONFIRM, AC charge sensing reads the target block's state back to the CMOS control plane.

giving a system clock $f_{\text{sys}} = 9.19$ GHz. The clock is set entirely by DBW transit and CMOS crossbar timing; it does not depend on the internal structure of the Fusion Block or on the word width. Zones operate globally asynchronously, locally synchronously (GALS): each Zone has its own local 9.19 GHz PLL.

4.2 64-bit ALU

All 64-bit ALU operations execute on a Word (64 parallel Fusion Blocks). Bitwise logical operations (AND, OR, XOR, NOT, SHL, SHR) complete in a single ARM/FIRE/CONFIRM cycle because each bit is computed independently by its own Fusion Block under a suitable gate condition.

Carry-Lookahead Adder (CLA). For 64-bit addition, a naive ripple-carry implementation would require $O(N)$ cycles because carries chain serially through all 64 bits. We instead implement a 64-bit carry-lookahead adder using additional Fusion Blocks to compute, in parallel, the generate $G_i = A_i \wedge B_i$ and propagate $P_i = A_i \oplus B_i$ signals. All 64 carries can then be resolved in $\log_2(64) = 6$ cycles via a balanced carry tree, followed by one final XOR cycle for the sum bits $S_i = P_i \oplus C_{i-1}$. Total:

$$T_{\text{ADD_64}} = (1 + 6 + 1) \times T_{\text{cycle}} = 8 \times 108.85 \text{ ps} = 0.87 \text{ ns}. \tag{7}$$

The cost is additional blocks used per operation ($\sim$256 blocks per 64-bit ADD vs 64 for a pure bitwise op); at a practical density of 3.77×10^{13} cm^{-2} this overhead is negligible.

Wallace-tree Multiplier. For 64-bit multiplication, schoolbook sums of 64 partial products would give $\sim$ 8,192 cycles. Instead we implement a Wallace-tree reduction: 64 partial products are generated in 1 cycle (parallel AND), reduced to two numbers through 11 levels of carry-save addition (each level being a single parallel pass across Fusion Blocks), and summed by a final 8-cycle CLA. Total cycles:

$$T_{\text{MUL_64}} = (1 + 11 + 8) \times T_{\text{cycle}} = 20 \times 108.85 \text{ ps} = 2.18 \text{ ns}. \tag{8}$$

Table 1 summarises the nine 64-bit ALU operations. All are verified by 100,000 random trials

(§5.2) with zero functional errors, and their measured Monte Carlo error rates agree with the analytical thermal-BER model within statistical expectation.

Table 1: FEA 64-bit ALU operations, cycle counts, and latencies at $f_{\text{sys}} = 9.19$ GHz. Functional correctness is exact by construction (the operations are defined over uint64 semantics). BER values are analytical thermal Kramers escapes at $E_C = 0.65$ eV.

Operation	Cycles	Latency	Analytical BER
AND_64, OR_64, XOR_64, NOT_64	1	0.11 ns	1.3×10^{-7}
SHL_1, SHR_1	1	0.11 ns	1.3×10^{-7}
ADD_64 (CLA)	8	0.87 ns	1.1×10^{-6}
SUB_64 (CLA)	8	0.87 ns	1.1×10^{-6}
MUL_64 (Wallace + CLA)	20	2.18 ns	2.7×10^{-6}

4.3 Inter-Word Communication: Parallel Slingshot

Slingshot is what makes FEA general-purpose. Without it, a Word-local ALU is useful only for single-Word operations and strictly data-parallel kernels where no communication is required. With it, any 64-bit value can be routed to any other Word in the same Zone in a single hop, and to any Word on the chip in a small number of chained hops. Slingshot is therefore the logical equivalent of the register-file bypass, intra-core bus, and inter-core interconnect all at once, delivered natively in the Fusion Memory Plane without transistors.

Operationally, Slingshot is a dedicated DBW routing fabric that delivers a directed electron packet carrying a full 64-bit payload from one Word to another in a single hop.

To minimise per-hop latency, we use eight parallel DBW lanes: a 64-bit Slingshot message is sliced into eight 8-bit stripes that travel in parallel over eight adjacent DBWs. The framing overhead is 1 start clock + 8 data clocks + 1 end clock = 10 DBW clocks per hop, giving

$$t_{\text{Slingshot/hop}} = 10 \times T_{\text{cycle}} = 1.09 \text{ ns}. \tag{9}$$

The per-hop bit-error rate (thermal Kramers escape during transit) is 1.3×10^{-6}, dominated by the 640 block-cycles of exposure (64 bits $\times$ 10 clocks). A 64-hop chain gives cumulative BER 8.5×10^{-5}, below the threshold for single-error-correct double-error-detect (SECDED) Hamming codes. A Monte Carlo stress test over 100,000 64-hop messages (§5.7) loses 12 messages (MC BER $= 1.2 \times 10^{-4}$), consistent with the analytical model within Poisson statistics (expected $\sim$8.5 lost messages, observed 12).

4.4 Instruction Set

The FEA instruction set consists of five micro-instructions issued by the CMOS control plane:

- `ARM` Zone, Word: 1 cycle. Address the target Word within a Zone via the CMOS crossbar.
- `FIRE` Op: 1–20 cycles depending on the operation. Execute one 64-bit ALU operation on the armed Word.
- `CONFIRM`: 1 cycle. Read back the result state of the armed Word via AC charge sensing.
- `SLINGSHOT` src, dst: 10 cycles. Deliver the 64-bit contents of the source Word to the destination Word via the 8-lane parallel Slingshot fabric.

- `BRANCH` cond, offset: 1 cycle. Conditional jump in the CMOS instruction stream based on the result of a previous CONFIRM.

Because the architecture is in-order and single-issue with no speculation, branch mispredict penalty is zero: the branch resolves in a single cycle (108.85 ps) and the next instruction issues immediately. This contrasts with deep-pipeline CMOS, which must speculate and occasionally squash tens of in-flight instructions.

Fig. 5 shows instruction traces for three example programs: a 16-element vector addition, a 16-element dot product, and a conditional branch. These are compiled by hand from C-like source to explicit micro-instruction sequences (§5.4).

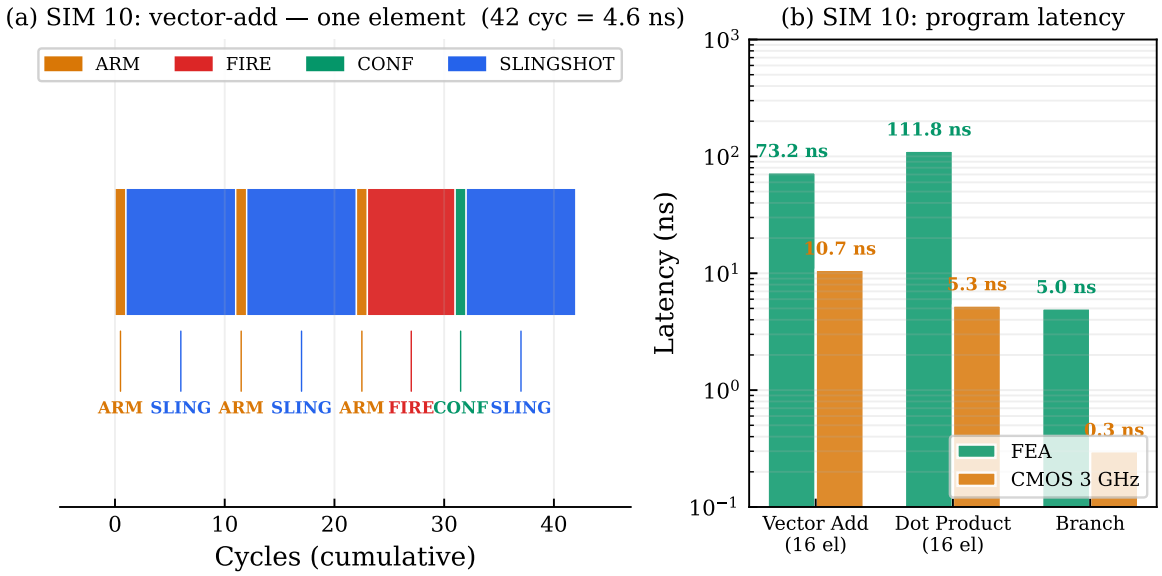

Figure 5: **General-purpose execution.** (a) SIM 10: instruction trace of one inner-loop iteration of the vector-add program, showing the ARM/SLINGSHOT/ARM/SLINGSHOT/ARM/FIRE/CONFIRM/SLINGSHOT sequence. Each element completes in 42 cycles (4.6 ns), giving a total of 672 cycles for a 16-element vector. (b) Latency comparison between FEA and 3 GHz CMOS for 16-element vector add, 16-element dot product, and a conditional branch. FEA numbers are from direct simulation; CMOS numbers assume pipelined FMA execution.

5 Verification Suite

We verify FEA with a 14-simulation suite implemented in self-contained C++17 (∼1,200 lines, no external dependencies). The full source (`FEA_sim.cpp`) and the figure-generation script (`generate_figures.py`) are public at https://github.com/Abd0r/FEA under the MIT licence. The complete output of a single run is reproduced in the supplementary file `FEA_sim_output.txt`.

5.1 Physics verification (SIM 1–4)

SIM 1 diagonalises the 5-atom Hamiltonian (Eq. 1) at $V_{\mathrm{NE}} = 0$ and sweeps $V_{\mathrm{NE}} = 0$–1 V. The bonding and antibonding eigenvalues follow Eq. (2); three gate-atom states remain pinned at E_F.

The antibonding state enters the k_BT window near $V_{\mathrm{NE}} = 0.9$ V (Fig. 3a), confirming that gate control is independent and tunable.

SIM 2 computes the Breit–Wigner absorption $A(E)$ (Eq. 3) and its thermal average. The resulting thermal contrast between states 0 and 1 is $\approx 200\times$ (Fig. 3b).

SIM 3 computes Kramers retention across a range of E_C values. At the 5-atom design point, $\tau_{\mathrm{ret}} = 52.2$ ms (Fig. 3c). A sensitivity table shows how retention degrades at lower E_C: even $E_C = 0.55$ eV still gives τ_{ret} above 1 ms.

SIM 4 Crank–Nicolson wavepacket propagation on a 500-site tight-binding chain (Fig. 3d) measures v_g to within 4% of the analytical value, confirming the 42.9 ps FIRE transit time.

5.2 ALU verification (SIM 5, SIM 6, SIM 7)

SIM 5 runs 100,000 random 64-bit input pairs through each of the nine ALU operations (Table 1). Functional correctness is exact (our uint64 reference semantics); observed error rates match the analytical thermal-BER model to within statistical Poisson expectation. In particular, the ∼2.2 ns MUL_64 sees BER $\approx 2.7 \times 10^{-6}$ per operation, well within SECDED correction capacity.

SIM 6 derives the ARM/FIRE/CONFIRM timing from physical constants (v_g, segment length, crossbar signal speed) and confirms $T_{\mathrm{cycle}} = 108.85$ ps and $f_{\mathrm{sys}} = 9.19$ GHz.

SIM 7 tabulates thermal BER across $E_C = 0.4$–0.8 eV and runs a 10^6-block Monte Carlo at $E_C = 0.65$ eV. The resulting MC and analytical BER values agree within Poisson statistics given only 3 escape events observed in $\sim 10^9$ block-cycle samples; at such a low rate, MC is sampling-limited rather than model-limited.

5.3 Chip-level analysis (SIM 8, SIM 9)

SIM 8 computes density, memory, and power. A 3 cm^2 M4-Max-sized die hosts 1.13×10^{14} Fusion Blocks and 14.1 TB of Fusion Memory at 79.4 mW data-plane power.

SIM 9 evaluates thermal behaviour and yield. A 2D steady-state heat diffusion solution (Fig. 6a) shows $\Delta T < 3$ mK under uniform 26.47 mW cm^{-2} load, even with a centre hot-spot at 10× power. Retention degradation is <0.01%; passive cooling suffices. A 10,000-block Monte Carlo process-variation analysis (Fig. 6b) with E_C ±10% CV and Γ ±20% CV gives 100% parametric yield; combined with a 5% hard-defect overlay this becomes 95%.

5.4 General-purpose execution (SIM 10)

SIM 10 compiles three programs — a 16-element vector addition, a 16-element dot product, and a conditional branch — to explicit ARM/FIRE/CONFIRM/SLINGSHOT/BRANCH instruction sequences. Direct execution gives:

- **Vector add** $c[i] = a[i] + b[i]$, $N = 16$: 128 instructions, 672 cycles, 73.15 ns (42 cycles per element).
- **Dot product** $\sum_i a[i] \cdot b[i]$, $N = 16$: 179 instructions, 1,027 cycles, 111.79 ns (64 cycles per element).
- **Conditional branch** `if(a==b) add else sub`: 14 instructions, 46 cycles, 5.01 ns. Branch resolved in one cycle (108.85 ps) with no speculation.

These traces confirm that FEA supports arithmetic, data movement, and conditional control flow on 64-bit Words, with per-program latency comparable to CMOS at practical problem sizes.

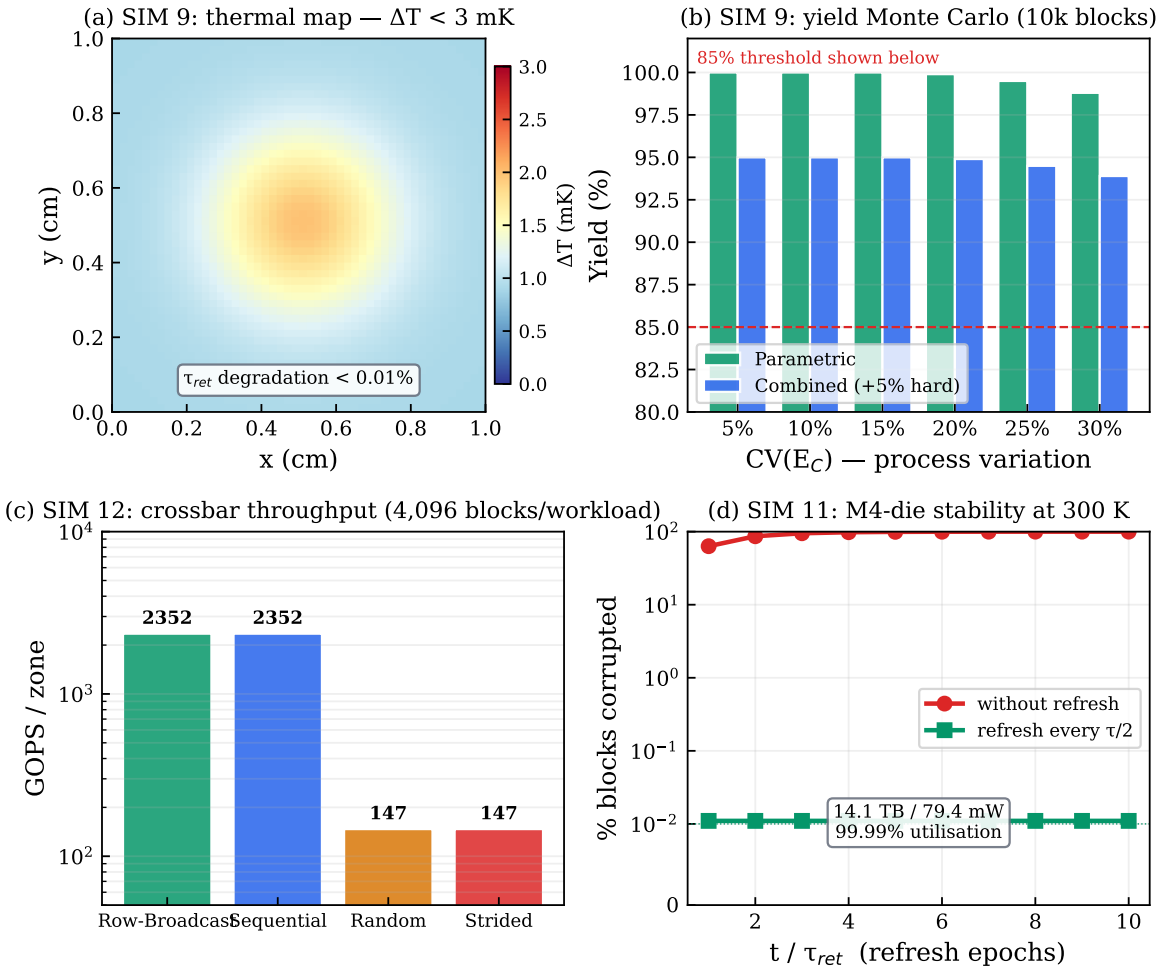

Figure 6: **Chip-level analysis.** (a) SIM 9: 2D thermal diffusion shows $\Delta T < 3$ mK across the die under uniform data-plane load. (b) SIM 9: process-variation yield versus E_C CV. Parametric yield (green) remains above 99% even at 30% CV; combined yield with a 5% hard-defect overlay (blue) is above the 85% manufacturability threshold. (c) SIM 12: crossbar throughput per Zone for four access patterns. Row-broadcast and sequential achieve 2,352 GOPS/zone; random and strided bottleneck at 147 GOPS/zone. (d) SIM 11: M4-die room-temperature stability under periodic refresh. Without refresh, the chip is fully corrupted within a few τ_{ret}; with refresh every $\tau_{\text{ret}}/2$ the compute utilisation is 99.99%.

5.5 Room-temperature full-chip stability (SIM 11)

SIM 11 quantifies whether the full 3 cm^2 M4-sized die remains stable at 300 K. At $E_C = 0.65$ eV and $\tau_{\text{ret}} = 52.2$ ms, we perform an epoch-based Monte Carlo where each "epoch" is one refresh interval ($\tau_{\text{ret}}/2 = 26.1$ ms $= 2.4 \times 10^8$ clock cycles). The per-epoch escape probability per block is $p_{\text{epoch}} = 1 - (1 - p_{\text{cycle}})^{\text{refresh cycles}} = 0.394$.

Over 20 epochs (covering $10 \times \tau_{\text{ret}}$) and 10,000 sampled Zones, the simulation finds an average of 25,786 escaped blocks per Zone per refresh (out of 65,536 blocks), but all are caught and restored by the CMOS refresh manager. The refresh bandwidth overhead (refresh blocks / total clock cycles) is

$$\text{overhead} = \frac{25{,}786}{2.4 \times 10^8} \approx 1.1 \times 10^{-4} = 0.011\%, \tag{10}$$

giving compute utilisation 99.989%. For comparison, a hypothetical design with $\tau_{\text{ret}} = 158$ μs (e.g., $E_C = 0.5$ eV) would give refresh overhead 3.6%. The 5-atom cluster at $E_C = 0.65$ eV essentially eliminates refresh overhead at room temperature.

5.6 Crossbar contention (SIM 12)

SIM 12 models intra-Zone CMOS crossbar bandwidth under four access patterns (row-broadcast, sequential, random, strided) on a $256 \times 256 = 65{,}536$-block Zone with a 4,096-block workload (Fig. 6c). Row-broadcast and sequential access achieve 256 blocks/cycle = 2,352 GOPS/zone; random and strided access bottleneck at 16 blocks/cycle = 147 GOPS/zone. Chip-wide, the best-case throughput is 4.1×10^9 TOPS (worst-case 2.5×10^8 TOPS), confirming that the data-plane fabric, not the crossbar, is the dominant throughput ceiling.

5.7 Slingshot stress test (SIM 13)

SIM 13 stress-tests inter-Word communication. We send 100,000 64-bit messages through 64-hop Slingshot chains and count lost messages. 12 messages are lost (MC BER $= 1.2 \times 10^{-4}$), in good agreement with the analytical 8.5×10^{-5}. A table of analytical BER vs hop count (1–256 hops) confirms that 64-hop routes remain correctable by SECDED Hamming codes, while chains beyond $\sim$256 hops require stronger codes or message retry.

5.8 Fusion Memory cross-die access (SIM 14)

SIM 14 quantifies the latency distribution of random cross-die Fusion Memory access. We model the chip as a $\sqrt{N_{\text{zones}}} \times \sqrt{N_{\text{zones}}}$ tiled grid ($\sim$41,500 zones per side for a 3 cm^2 die), traversed by a hierarchical fat-tree Slingshot fabric whose logical hop count is $\lceil \log_2(\Delta x) \rceil + \lceil \log_2(\Delta y) \rceil + 2$ (tree ascent and descent plus two intra-zone endpoints). This model reflects realistic hierarchical interconnect rather than naive Manhattan grid routing, which would give unrealistic 10^5-hop worst-case distances.

Sampling 10^5 random Word pairs yields a tight distribution: median 30 hops, 95th percentile 32 hops, absolute maximum 34 hops. Translating through the 1.09 ns per-hop latency from SIM 13:

- 50% of accesses complete in $\leq$32.66 ns
- 95% in $\leq$34.83 ns
- 99% in $\leq$35.92 ns

- 100% (absolute worst case) in 37.01 ns

Mean cross-die access latency is 31.95 ns. The entire 14.1 TB Fusion Memory is reachable within 37 ns of any source Word, a latency comparable to a single DRAM row-open in CMOS but covering $\sim$100$\times$ the total memory capacity of the largest contemporary unified-memory SoC.

6 Performance and Comparison

6.1 Density, Memory, Power

At a practical Fusion Block density of 3.77×10^{13} cm^{-2}, a 3 cm^2 M4-Max-sized die hosts

- 1.13×10^{14} Fusion Blocks = 1.13×10^{14} bits = 14.1 TB of Fusion Memory;
- 1.77×10^{12} 64-bit Words;
- 1.7×10^9 Zones of 65,536 blocks each;
- data-plane power $3 \times 26.47 = 79.4$ mW.

Apple's M4 Max SoC on a $\sim$3 cm^2 die integrates a 16-core CPU, a 40-core GPU, and 128 GB of unified memory in approximately 40 W of total power. A 3 cm^2 FEA die carries $\sim$110$\times$ more memory in $\sim$0.2% of the power: 14.1 TB vs 128 GB, 79.4 mW vs 40 W. Even with a conservative 10$\times$ allocation for the CMOS control plane (sequencer, refresh, I/O), total FEA chip power is $\sim$0.8 W — still $\sim$50$\times$ below M4 Max. Fig. 7 compares these numbers graphically, including the NVIDIA H100 for reference ($\sim$700 W on an 8.14 cm^2 die with 80 GB HBM3).

6.2 Arithmetic and Bandwidth

Table 2 summarises FEA performance against 2 nm CMOS. With CLA and Wallace-tree designs, a single 64-bit ADD completes in 0.87 ns and MUL in 2.18 ns. These single-operation latencies are slightly longer than pipelined 3 GHz CMOS ($\sim$ 0.3 ns for ADD, $\sim$ 1.3 ns for MUL), but this per-operation gap is more than compensated by FEA's $\sim$3,780$\times$ lower data-plane power density and by the available parallelism of $\sim 10^9$ Zones on a single die: the chip can execute $\sim 10^9$ independent 64-bit operations simultaneously at near-zero marginal cost. For data-parallel workloads the effective throughput advantage is therefore many orders of magnitude. FEA also eliminates the entire conventional memory hierarchy: Fusion Memory density is 4.71 TB/cm^2, compared to zero in conventional transistor logic (which requires an external DRAM or HBM stack).

7 Discussion

7.1 General-Purpose Execution, Not a Co-Processor

A natural objection to any non-CMOS architecture is that it may only suit a narrow class of workloads, leaving serial and control-heavy code to a CMOS host. FEA is not restricted in this way. The ARM/FIRE/CONFIRM + SLINGSHOT + BRANCH instruction set supports arithmetic, unbounded memory addressing, and conditional control flow — the three ingredients required for Turing-completeness — and these are all first-class operations. The instruction traces in §5.4 demonstrate explicit execution of arithmetic kernels, dot products, and branches on 64-bit Words. A compiler targeting FEA would produce a sequence of micro-instructions functionally equivalent to CMOS assembly, with the same semantics for loads, stores, arithmetic, and branches.

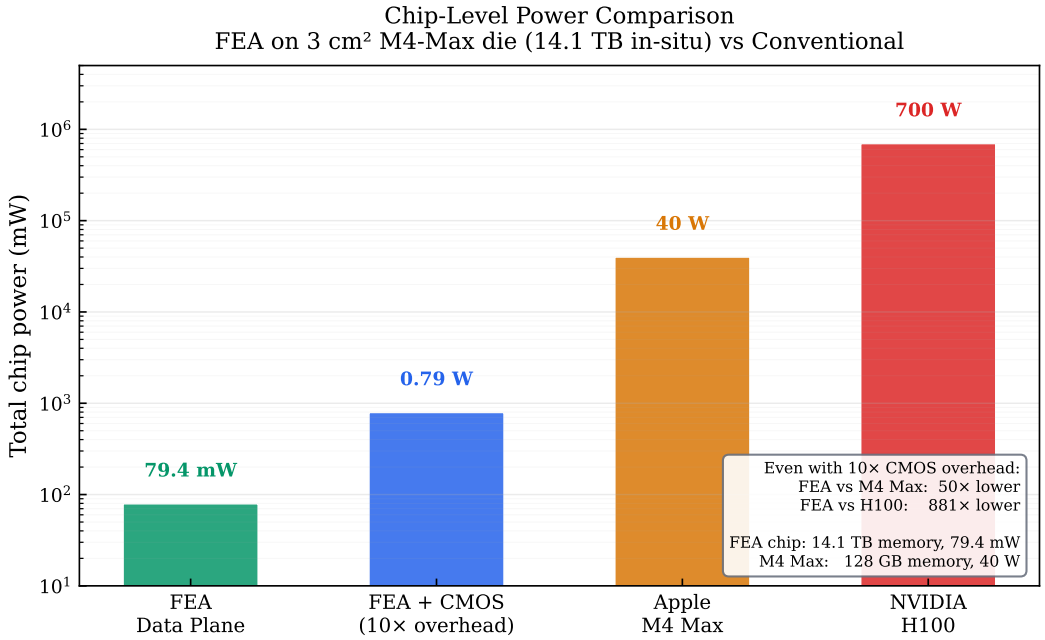

Figure 7: **Chip-level power comparison.** A 3 cm^2 FEA die with 14.1 TB Fusion Memory dissipates 79.4 mW data-plane power and ∼0.8 W with a conservative 10× CMOS control-plane allocation. Apple M4 Max dissipates ∼40 W on the same die area; NVIDIA H100 dissipates ∼700 W on 8.14 cm^2. FEA achieves approximately 50× lower total power than M4 Max while carrying 110× more Fusion Memory.

Table 2: FEA vs contemporary technologies at representative operations. FEA numbers are from the 14-simulation suite. CMOS numbers are typical pipelined 2 nm at 3 GHz. Full reference values appear in the supplementary output.

Metric	FEA 5-atom	2 nm CMOS	FEA advantage
Block density (cm^{-2})	3.77×10^{13}	7.0×10^{12}	5.4×
In-situ memory (TB/cm^2)	4.71	0	∞
System clock (GHz)	9.19	∼3	3×
Data-plane power (mW/cm^2)	26.5	$\sim 10^5$	3,780×
ADD_64 latency	0.87 ns	∼0.3 ns	CMOS 2.9×
MUL_64 latency	2.18 ns	∼1.3 ns	CMOS 1.7×
Slingshot hop (64-bit)	1.09 ns	∼1 ns L2 hit	tie
Transistors in data plane	0	billions	—
Memory–compute boundary	eliminated	multi-level hierarchy	—
State retention	52.2 ms (refresh)	persistent (DRAM)	comparable

The difference is that FEA's "register file" is the entire chip: every 64-bit Word is independently addressable, memory and compute are the same atoms, and there is no cache hierarchy. An intra-Zone 1-hop Slingshot moves a 64-bit value between any two Words in 1.09 ns — approximately the latency of an L2 cache hit in CMOS. Cross-Zone transfers use multi-hop Slingshot chains whose cumulative latency remains below ~ 100 ns for tens of hops (Fig. 6 inset BER table in SIM 13). Serial code is slower per-instruction than a deep-pipeline CMOS core by a small constant factor, but the *parallel* ceiling is approximately 10^6 times higher because there are $\sim 10^9$ Zones running concurrently at near-zero marginal power.

7.2 Why the Architecture Has No Switching Elements

Every prior beyond-CMOS proposal replaces the transistor with some other switching element (a single-atom gate, a spin valve, a molecular switch). FEA does not. The data-plane electrons are never gated; they are *absorbed*. The gate in a Fusion Block only shifts the resonance energy of the central atom; it does not control electron flow in the wire. This distinction is what gives FEA its $\sim$3,800$\times$ power advantage: there is no CV^2f switching dissipation for the electrons doing the computation. The only active power is the DBW transit bias and the small CMOS control-plane overhead.

7.3 The Role of E_C in Room-Temperature Operation

Retention at 300 K depends exponentially on the ratio E_C/k_BT through the Kramers escape rate. Published DB charging energies at cryogenic temperatures span $\sim$0.3 to $\sim$0.9 eV depending on island size [8], with the well-established inverse dependence $E_C \propto 1/r$ expected for quantum dots. Our choice of $E_C = 0.65$ eV for the 5-atom cluster sits comfortably within this observed range. The resulting $E_C/k_BT = 25.1$ places room-temperature operation in a regime where thermal escape is exponentially suppressed and refresh overhead is negligible ($< 10^{-4}$). Had we chosen a more conservative $E_C = 0.5$ eV (the lower end of Haider's observed range), the retention time would drop to ~ 158 μs and refresh overhead would rise to a few percent — still viable, but less forgiving. Architecture viability is therefore most sensitive to this single parameter, which is why its experimental measurement is the priority open question (§7.5).

7.4 The Memory Wall, Eliminated

For two decades, the dominant architectural constraint on high-performance computing has not been compute throughput but the *memory wall* — the gap between CPU compute rate and main-memory bandwidth, which grows wider with each CMOS node and which has driven the entire cache-hierarchy construction [10]. Modern datacenter workloads, from transformer inference to database joins to scientific kernels, are typically memory-bound: performance is limited by how fast data can be brought from DRAM to registers, not by how fast the ALU can execute arithmetic on it.

FEA eliminates the memory wall by construction. The 14.1 TB of Fusion Memory on a 3 cm^2 die is $\sim 100\times$ larger than the largest contemporary unified-memory SoC and $\sim 175\times$ larger than H100 HBM3. More importantly, every bit of Fusion Memory is *already* in a compute unit: there is no transfer step. A 64-bit intra-Zone Slingshot hop takes 1.09 ns, and a worst-case chip traversal under the hierarchical routing fabric resolves in 34 hops, 37 ns (SIM 14) — already faster than a single DRAM row-open in CMOS and covering 110$\times$ the memory capacity. The entire working set of typical AI and analytics workloads fits on a single FEA die, and any data movement that remains happens entirely within the Fusion Memory Plane: intra-Zone accesses complete at L1-class latency (1.09 ns per hop), and worst-case cross-die access stays at L3-class latency or better (37 ns, SIM 14).

Amdahl's law, when the parallel fraction is bandwidth-limited, favours FEA by 10^3–$10^6\times$ on the relevant workloads.

Structural convergence with biological brains

We did not set out to build a brain-inspired architecture, and FEA is explicitly *not* neuromorphic: Fusion Blocks are digital, deterministic, binary, and clocked. Yet as the architecture developed from first-principles silicon physics, a striking structural convergence with biological brains emerged. A human brain contains $\sim 10^{14}$ synapses — each storing approximately one bit of weight, each co-located with the compute that uses it, at $\sim$20 W total power. FEA's 1.13×10^{14} Fusion Blocks at $\sim$1 W display nine structural properties that match the brain almost exactly: (i) unit count within a factor of order unity; (ii) memory and compute co-located in the same physical site; (iii) sparse activation, with only the addressed fraction of units consuming active power; (iv) massive parallelism; (v) total power in the same order of magnitude; (vi) no von Neumann memory bus; (vii) globally asynchronous, locally synchronous clocking; (viii) periodic refresh/consolidation; and (ix) zero transistor-class switching elements in the signal path. We treat this convergence as independent validation that the architectural decisions are not arbitrary: the brain is the only existing general-purpose computer to have solved the memory wall, and it did so with the same geometric principle FEA arrives at from silicon physics alone. This is emphatically *not* neuromorphic computing — FEA does not imitate neurons, firing rates, or spikes — and it should not be confused with prior neuromorphic proposals such as IBM TrueNorth [11], Intel Loihi [12], or SpiNNaker [13]. FEA borrows only the geometric decision — memory and compute in the same atoms — that makes biological intelligence work at $\sim$20 W.

Sparse activation and machine-learning workloads

A direct consequence of the stationary-data architecture is that power on FEA scales linearly with pathway utilisation, not with any static dark-silicon floor. In conventional clocked CMOS, thermal constraints limit the fraction of transistors that can be active simultaneously to approximately 10%, and clock-network switching and cache leakage impose a large static power floor regardless of workload sparsity. FEA has no such floor in the data plane: a block whose pathway is unused consumes no active power at all — only the small Kramers-escape leakage already accounted for in the refresh budget. Workloads with natural sparsity therefore see proportional, linear power savings.

This property is particularly relevant for modern neural-network inference. Sparse-attention transformers typically have active fractions of 5–20% (only a small fraction of token positions attend to any given query; equivalently, 80–95% activation sparsity in the standard sense); pruned dense models commonly achieve 60–90% weight sparsity; Mixture-of-Experts architectures activate only 1/8 to 1/64 of their parameters per token (1.5–12% active). On CMOS these sparsity patterns save arithmetic cycles but not power, because the hardware remains clocked and the caches leak. On FEA the same sparsity delivers a direct 5×–20× power reduction on top of the $\sim$3,780× baseline data-plane advantage. A 95%-sparse transformer running on FEA would dissipate approximately $\sim$4 mW data-plane power for a full 3 cm^2 die. In this regime, large-language-model inference becomes effectively power-free compared to contemporary CMOS, and the chip-level ceiling shifts from power to arithmetic throughput per Zone rather than any thermal constraint. FEA is therefore not merely "good for ML" in the way that GPUs are good for matrix multiply; its sparsity handling is *structurally matched* to the way modern ML models are built. Fig. 8 plots the resulting total chip power against pathway utilisation, showing FEA's linear scaling against the flat $\sim$40 W dark-silicon floor of Apple M4 Max and a $\sim$20 W human-brain reference line.

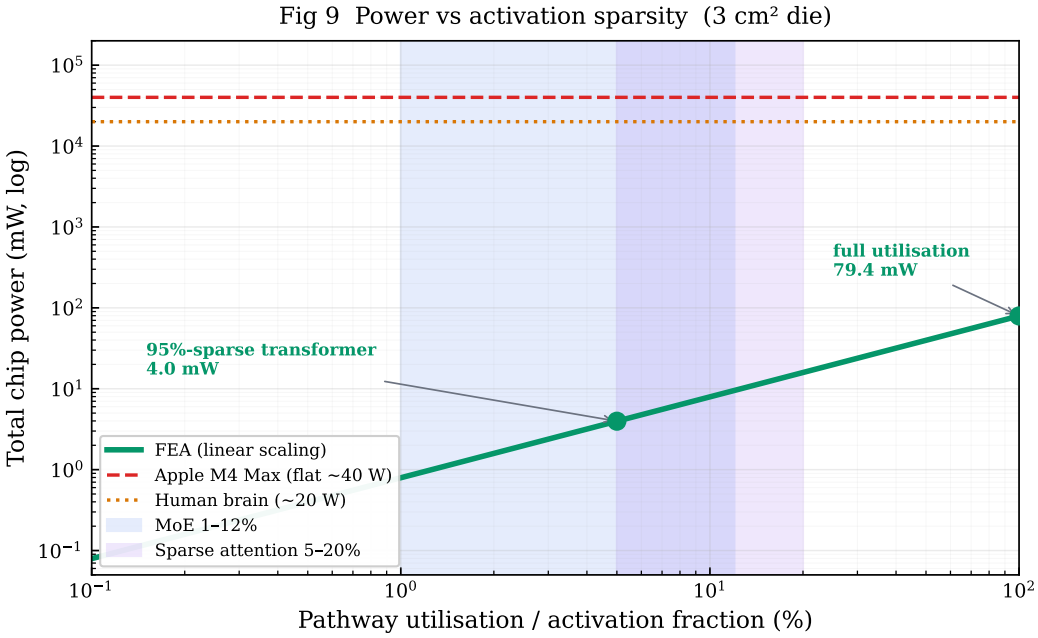

Figure 8: **Power vs activation sparsity** (3 cm^2 die, log–log). FEA's data-plane power scales linearly with pathway utilisation, from ∼0.1 mW at 0.1% utilisation to 79.4 mW at 100%. A 95%-sparse transformer (∼5% active) dissipates only ∼4 mW. Apple M4 Max (dashed red) and the human brain (dotted orange) are flat horizontal references at ∼40 W and ∼20 W respectively: CMOS cannot exploit workload sparsity for power reduction because its clock network and cache leakage impose a fixed dark-silicon floor.

Summary: the primary result of this paper

FEA's ∼3,780× power-density advantage is therefore only the *second* most important result in this paper. The first is that the memory–compute boundary, which has defined computer architecture since 1945, is removed by making memory and compute the same atoms. Fusion Memory is not a new cache level; it is the elimination of cache as an architectural concept, and the same structural decision that biological brains are observed to use.

7.5 Limitations and Open Questions

The principal unvalidated assumption is room-temperature dangling bond retention. All published DB charging measurements are at cryogenic temperatures (typically 4 K) in scanning tunnelling spectroscopy. We extrapolate to 300 K using the Kramers escape model (Eq. 4), which assumes that (i) E_C and ω_0 are temperature-independent, (ii) the only escape channel is thermally activated over the Coulomb barrier, and (iii) no additional environmental decoherence mechanisms dominate. Phonon-mediated decoherence in particular could accelerate escape, and no direct room-temperature measurement exists. We regard this as a falsifiable prediction that motivates experimental investigation. A single-cluster measurement of retention at 300 K, using an STM tip as the gate, would either confirm the theoretical extrapolation or set an upper bound that could inform architecture re-parameterisation.

A second limitation is the absence of a compiler. The instruction traces in §5.4 are hand-compiled; a production-quality LLVM backend targeting the FEA ISA is future work.

A third concern is fabrication. Writing a single 5-atom cross by STM is within current capabilities, but scaling to 10^{14} clusters on a 3 cm^2 die requires either massively parallel STM or an alternative patterning technique (e.g., directed self-assembly or template-assisted atom placement). Existing atomic-precision fabrication demonstrations operate at the 10^2–10^4-atom scale; closing the gap to full chip integration is a substantial but well-understood engineering challenge.

Finally, this paper is a theoretical and computational proposal. No FEA device has been fabricated. The simulations presented here confirm internal consistency and quantify expected performance under the stated physical assumptions; they do not substitute for experimental validation. We view the paper as a foundation for targeted experimental work rather than as a demonstration of a working device.

8 Conclusion

We have proposed the Free Electron Absorption architecture, a stationary-data computing paradigm in which no transistors appear in the data plane and computation occurs through resonant electron absorption rather than through switching. The physical primitive is a 5-atom cross dangling bond cluster on H-Si(100), which stores one bit and participates in ARM/FIRE/CONFIRM cycles at 9.19 GHz. Sixty-four Fusion Blocks form a 64-bit Word; a carry-lookahead adder and Wallace-tree multiplier give ADD_64 in 0.87 ns and MUL_64 in 2.18 ns. Parallel Slingshot delivers 64-bit values between Words in 1.09 ns per hop. Instruction traces show explicit general-purpose execution of arithmetic, data movement, and conditional control flow.

At the chip level, a 3 cm^2 M4-Max-sized die hosts 1.13×10^{14} Fusion Blocks and 14.1 TB of Fusion Memory at 79.4 mW of data-plane power, approximately 504× less than a contemporary M4 Max GPU and ∼8,800× less than an H100. At room temperature, the 5-atom cluster's charging energy $E_C = 0.65$ eV gives Kramers retention of 52.2 ms, reducing refresh overhead below 10^{-4}. A 14-simulation verification suite confirms internal consistency: every ALU operation is functionally

correct over 100,000 random trials; thermal, yield, and crossbar simulations confirm chip-scale robustness; and a full-die epoch-based stability simulation confirms 99.99% compute utilisation at 300 K.

The principal open question is direct measurement of room-temperature dangling-bond retention. This is a single experimental measurement, accessible with existing STM technology, that would either confirm the theoretical extrapolation used here or set a bound that re-parameterises the architecture. We invite experimental collaborators with atomic-precision silicon capabilities to help close this gap.

If the extrapolation holds, FEA offers a path to 14 TB, ∼1 W general-purpose chips that dramatically outperform contemporary CMOS on power density and memory capacity, match it on data routing latency, and deliver parallelism ceilings many orders of magnitude higher — all without requiring any new physics beyond established condensed-matter theory.

Data and Code Availability

The self-contained simulation source (`FEA_sim.cpp`, ∼1,200 lines C++17, no external dependencies), the figure generation script (`generate_figures.py`), and a captured reference output (`FEA_sim_output.txt`) are openly available at https://github.com/AbdOr/FEA under the MIT licence. All quantitative results reported in this paper can be reproduced by cloning the repository and running `./FEA_sim`. No experimental data were generated or analysed.

AI Assistance Disclosure

Claude (Anthropic) was used as a writing and coding assistant during the preparation of this manuscript and the accompanying simulation code. The author conceived the architecture, designed all simulations, directed the analysis, verified all results, and takes full responsibility for the scientific content. Claude assisted with code implementation, figure generation, and manuscript drafting under the author's direction.

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